

# Ayush Ravi Chandran

ayushravichandran@gmail.com | [linkedin.com/in/ayush-ravichandran](https://www.linkedin.com/in/ayush-ravichandran) | [github.com/espiobest](https://github.com/espiobest) | [espiobest.me](https://espiobest.me)

## EDUCATION

### University of Massachusetts Amherst

Master of Science in Computer Science

Sep 2026 – May 2027

Bachelor of Science in Computer Science, Bachelor of Science in Mathematics GPA: 3.86/4.00 | Sep 2022 – May 2026

- **Coursework:** Neural Networks, Systems for Machine Learning, Algorithms for Data Science, Distributed Systems, Graduate Machine Learning and Artificial Intelligence
- **Awards:** Bay State Fellowship (Full Tuition, M.S.), Chancellor's Scholarship (40% Tuition, B.S.), Dean's International Scholarship, Flynn Research Scholarship, Dean's List, Outstanding TA Award
- **Teaching Assistant:** CS 240 Reasoning Under Uncertainty – led discussions, held office hours, and graded assignments for 100+ students

## EXPERIENCE

### Software Engineering Intern

September 2025 - Present

Center for Data Science and AI - University of Massachusetts Amherst

Amherst, MA

- Built scalable data pipelines (**PostgreSQL**, **S3**, **Azure**) to ingest and normalize multi-organization salt marsh data for a **Massachusetts DEP**-funded initiative, with schema validation and per-record error handling
- Owned end-to-end full-stack development of a multi-tenant web portal (**React**, **FastAPI**), applying **dependency injection** to separate service-layer logic, serving **1,000+** weekly field records across **20+** organizations
- Built a cross-platform mobile app (**Flutter**) with offline-first data capture, local **SQLite** storage, nested survey workflows, and automated QC validation, replacing a third-party tool for **100+** field researchers
- Serve as sole developer on the Salt Marsh Data Platform in production, writing, testing, and deploying code while directly communicating with **100+** field researchers to resolve issues

### Software Engineering Intern

May 2025 - August 2025

Massachusetts Green High Performance Computing Center (MGHPCC) - Unity HPC Cluster

Holyoke, MA

- Built a **Python** data analysis and reporting module to process job logs across the Unity HPC cluster, prototyping **Grafana** dashboards for real-time observability across **12+** months
- Optimized **SQL** queries and **data indexing** strategies to accelerate SLURM log analysis by **80%**, enabling faster anomaly detection across **10.9M+** job records
- Applied **unsupervised clustering** (DBSCAN) to identify 63,000+ underutilized NVIDIA A100 GPU hours, uncovering **\$120K+** in wasted compute resources

## PROJECTS

### Commit Bingo AI | TypeScript, Hono, Cloudflare Workers, Gemini API | [bingo.espiobest.me](https://bingo.espiobest.me)

July 2026

- Built a shipped **LLM**-powered developer analytics tool integrating the **Google Gemini API**, enforcing strict **structured JSON** output via prompt design rather than free text
- Implemented a **circuit breaker** pattern with sequential model fallback, gracefully degrading to local heuristics under rate limits to maintain reliability in production
- Deployed on **Cloudflare Workers** with secure **GitHub OAuth** using HTTP-only cookies

### Poker Bot | Python, PyTorch, NumPy | [poker-gui.vercel.app](https://poker-gui.vercel.app)

April 2025

- Built a **Deep Q-Network** reinforcement learning agent in **PyTorch** to play Heads-Up Limit Texas Hold'em, designing probabilistic game-state encoding using **Monte Carlo** win-rate estimates
- Trained agent via **self-play** with experience replay, achieving **4th/23** ranking and demonstrating emergent strategies like optimal folding

### RouteAble | TypeScript, NestJS, PostgreSQL, React Native, PyTorch | [github.com/RouteAble](https://github.com/RouteAble)

November 2023

- Launched a full-stack app in **36 hours** to crowdsource images of inaccessible locations, enhancing navigation for users with mobility challenges
- Developed a **PostgreSQL** backend with Dockerized **ML models** for obstacle detection and similarity, achieving **92%** accuracy in identifying accessibility barriers
- Received **Most Impactful Award** (\$2,000 prize, UChicago Winter Tech Showcase '24) and **Best Use of GitHub** (HackUMass, **Major League Hacking**)

## SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Go, C, C++

**Frameworks:** React, Next.js, FastAPI, Node.js, Express.js, NestJS, Flutter, PyTorch, TensorFlow, Pandas, scikit-learn

**Developer Tools:** PostgreSQL, Docker, Kubernetes, AWS (S3, Lambda, DynamoDB), Azure, Redis, Grafana, SLURM, Git, Linux